Hospital Patient Record System with Data Management Using Structures and pointer-Based Access

COURSE CODE: CSA0203

COURSE NAME: C Programming

SUPERVISOR :
Dr.S.K . Saravanan

PRESENTED BY:
Vennappa Gari Vijay
A Guna Sekar
P V Rahul Kavi

Introduction

- Hospital management requires efficient handling of patient records.
- Manual record keeping is time-consuming and prone to errors.
- This project is developed using the **C programming language**.
- It uses **structures** to store patient information.
- **Pointers** are used for efficient data access and management.

Problem Statement

- Hospitals maintain a large number of patient records daily.
- Manual record management is slow and inefficient.
- Searching patient information takes more time.
- Billing calculations may contain human errors.
- Paper records can be damaged or misplaced.
- A computerized system is needed for better management

Abstract

- This project manages hospital patient information digitally.
- It is implemented using **structures and pointers in C**.
- It stores registration, medical history, treatment, and billing details.
- It reduces paperwork and manual effort.
- It improves data retrieval and record maintenance.

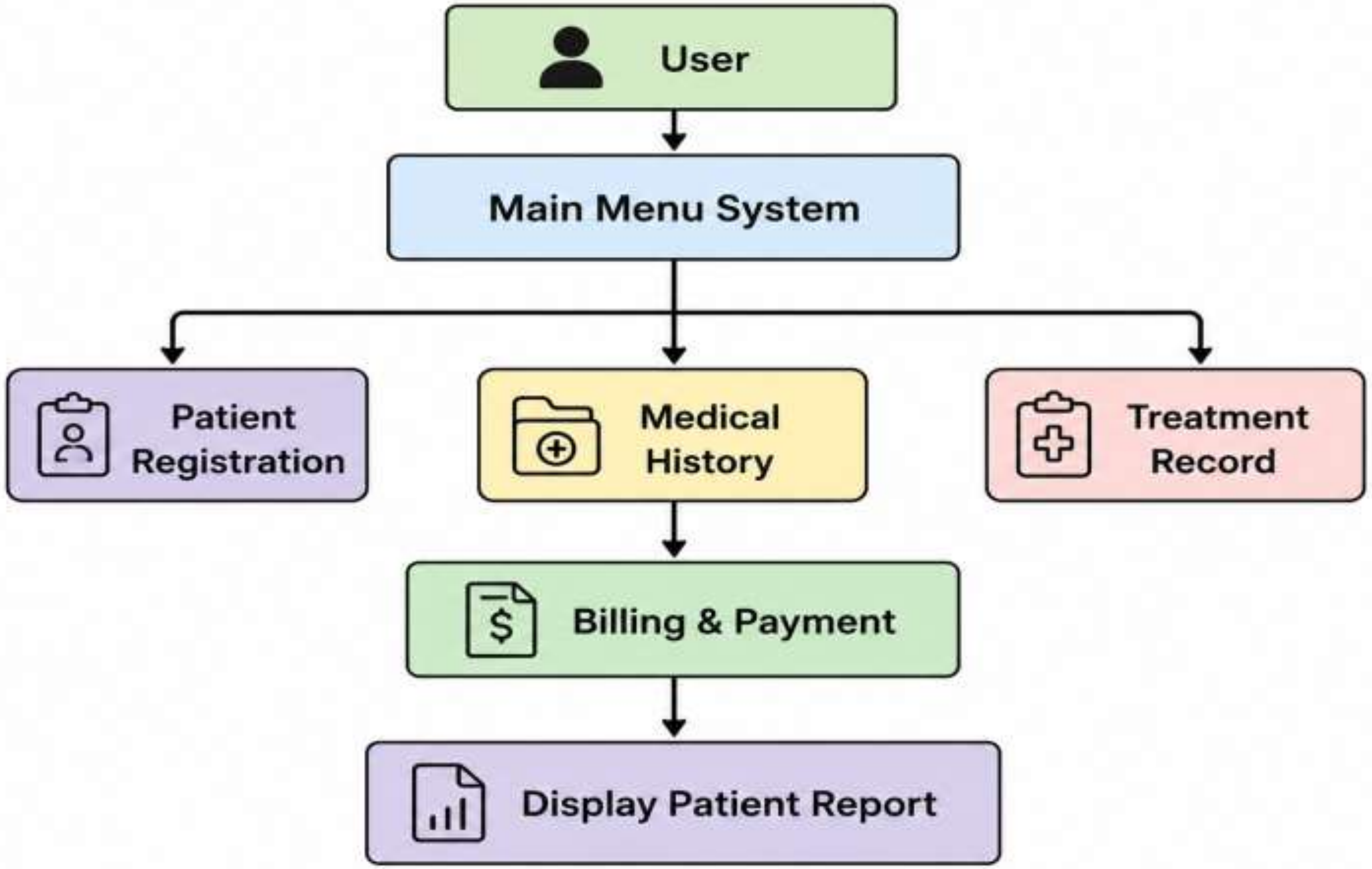
Existing System

- Patient records are maintained manually
- Billing is calculated manually.
- Record searching is difficult.
- Large amount of paperwork is required.
- Data duplication may occur.
- Human errors reduce efficiency.

Proposed System

- Computerized patient record management system.
- Uses structures for storing patient information.
- Uses pointers for efficient data access. Maintains
- patient, treatment, and billing records.
- Generates accurate billing automatically.
- Provides faster and secure record management..

Architecture



Conclusion & Future Enhancements

Conclusion

➢Successfully manages hospital records.

➢Uses structures and pointers effectively.

➢Improves hospital data management.

Future Enhancements

➢File handling for permanent storage.

➢MySQL database integration.

➢Login authentication system.

REFERENCES

- E. Balagurusamy, *Programming in ANSI C*, McGraw Hill.
- Brian W. Kernighan & Dennis M. Ritchie, *The C Programming Language*.
- Yashavant Kanetkar, *Let Us C*, BPB Publications.
- GCC Compiler Documentation.
- Code::Blocks / Dev-C++ Documentation.
- Software Engineering textbooks and class notes.

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

THANK YOU !!